

Annette Tongsak

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EDUCATION

Oregon State University, Corvallis, OR Sept 2022 - June 2026
B.S. Computer Science - Applied Graphics Simulation & Game Programming GPA: 3.93
Relevant Coursework: Computer Graphics & Shaders, Linear Algebra, Parallel Programming, Operating Systems, Software Engineering, Analysis of Algorithms, Databases, Deep Learning, Machine Learning

EXPERIENCE

Crowds Technical Director Intern June 2025 - August 2025
Pixar Animation Studios, Emeryville, CA

- Supported production shots in *Toy Story 5* by integrating large-scale crowd simulations and character animation assets into final rendered sequences within Pixar's animation pipeline
- Developed Python-based pipeline tooling using internal APIs to automate crowd character pairing workflows, reducing manual setup time by ~90% and improving workflow consistency for artists
- Set up, simulated, and debugged crowd scenes, ensuring stability and compatibility across diverse character assets and simulation configurations in collaboration with animators and technical teams
- Created reusable character animation assets and animator-facing rig controls to streamline scalable crowd workflows

Pixar Undergraduate Program Technical Director Intern June 2024 - August 2024
Pixar Animation Studios, Emeryville, CA

- Rotated across pipeline departments including set dressing, layout, modeling, shading, lighting, rigging, and effects using proprietary and industry-standard tools including Maya, Houdini, Katana, Nuke, RenderMan, and USD
- Collaborated on a team-produced animated short, contributing animation work and developing a Blender-based facial rigging system to automate texture switching and streamline the animation workflow
- Developed a procedural animation workflow using Houdini (VEX) and Blender to generate character motion from user-defined curves, reducing manual keyframing requirements

Undergraduate Research Assistant June 2023 - May 2024
Oregon State University, Corvallis, OR

- Conducted research in computer graphics, data visualization, and machine learning under Dr. Yue Zhang
- Investigated graphics techniques including path tracing, multithreading, vectorization, and optimization methods
- Contributed to a wildlife object detection project for the Oregon Department of Transportation through dataset preparation and model development

PROJECTS

Auto Lip Sync Blender Add-on | *Python, Blender API, WhisperX, Phonemizer* January 2026 - Present

- Developing a Blender add-on that automates first-pass lip sync generation through speech-to-phoneme alignment, producing animator-editable facial animation keyframes
- Built artist-facing workflow tools for viseme mapping, validation, and animation generation within Blender
- Researched and evaluated phoneme alignment approaches to improve timing accuracy and production usability
- Implemented support for multiple viseme standards, CPU/GPU execution, and non-destructive animator workflows

Random Cobweb Generator | *C/C++, OpenGL* December 2023

- Developed a procedural cobweb generation system inspired by DreamWorks' 2011 paper "Building and Animating Cobwebs for Antique Sets"
- Applied mathematical modeling of catenary curves and parabolas to generate 3D web structures

SKILLS

Programming: Python, C/C++, JavaScript, SQL

Animation & Pipeline: Maya, Houdini, Blender, USD, Katana, Nuke, RenderMan

Pipeline Development: Linux/Unix, Git, Perforce, Docker

Graphics & Performance: OpenGL, GLSL, CUDA, OpenCL, OpenMP, MPI

Developer Tools: Jira, Postman, Visual Studio, VS Code